



Stony Brook Institute
at Anhui University

安徽大学纽约石溪学院

ISE 333: User Interface Development (Design)

Chapter 7: Direct Manipulation and Immerse Environments

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Direct Manipulation

- Conception
 - Most users today are used to a *What You See Is What You Get* (WYSYG) environment enhanced by direct-manipulation methods
- Three Principles of Direct Manipulation
 - Continuous representations of the objects and actions of interest with meaningful visual metaphors
 - Physical actions or presses of labeled UI objects instead of complex syntax
 - Rapid, incremental, reversible actions whose effects on the objects of interest are visible immediately

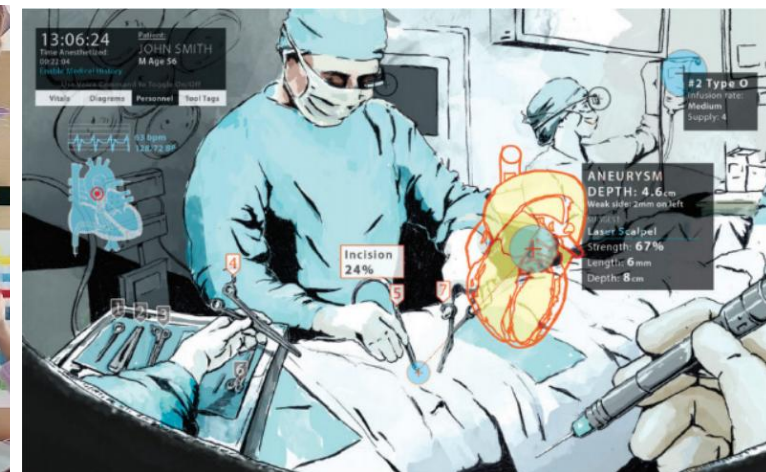
Direct Manipulation

- UI Metaphors
 - Practices of mapping familiar real-world concepts (source domain) onto digital entities (target domain) to create an intuitive experience
 - Desktop: Treats the screen as a physical desk with folders, files, a trash can
 - Web portal/page: Similar to a magazine or book page
 - File folders/tree view: Represents file hierarchies
 - Shopping cart: Used in e-commerce to mimic in-store shopping
 - Toolbar/menu bar: Simulates a set of tools on a workbench
 - Using familiar metaphors creates easier learning conditions for users and lessens the number of mistakes and incorrect actions



Direct Manipulation

- Strength / Translational Distance
 - Strength of the direct-manipulation metaphor, i.e. translational distance between users and the representation of the metaphor, can be perceived along a continuum from weak to immersive
 - Weak Early video game controllers
 - Medium Touchscreens, multi-touch
 - Strong Data glove, gesturing, manipulating tangible objects
 - **Immerse** VR (virtual reality), AR (augmented reality)



Direct Manipulation

- Advantages (Pros) of Direct Manipulation
 - Vividly presents task concepts
 - Allows easy learning, easy retention, more likely avoidance of errors
 - Encourages exploration
 - Affords high subjective satisfaction
- Disadvantages (Cons) of Direct Manipulation
 - May be hard to program
 - Is a setback for vision-impaired users
 - Consumes valuable screen space, requires scrolling or multiple actions

Examples of Direct Manipulation

- Geographical Systems
 - Bei Dou / Americal GPS satellite navigation system
 - Baidu / Google street views



Examples of Direct Manipulation

- Video Games
 - Mine sweeper
 - Super Mario
 - World of warcraft



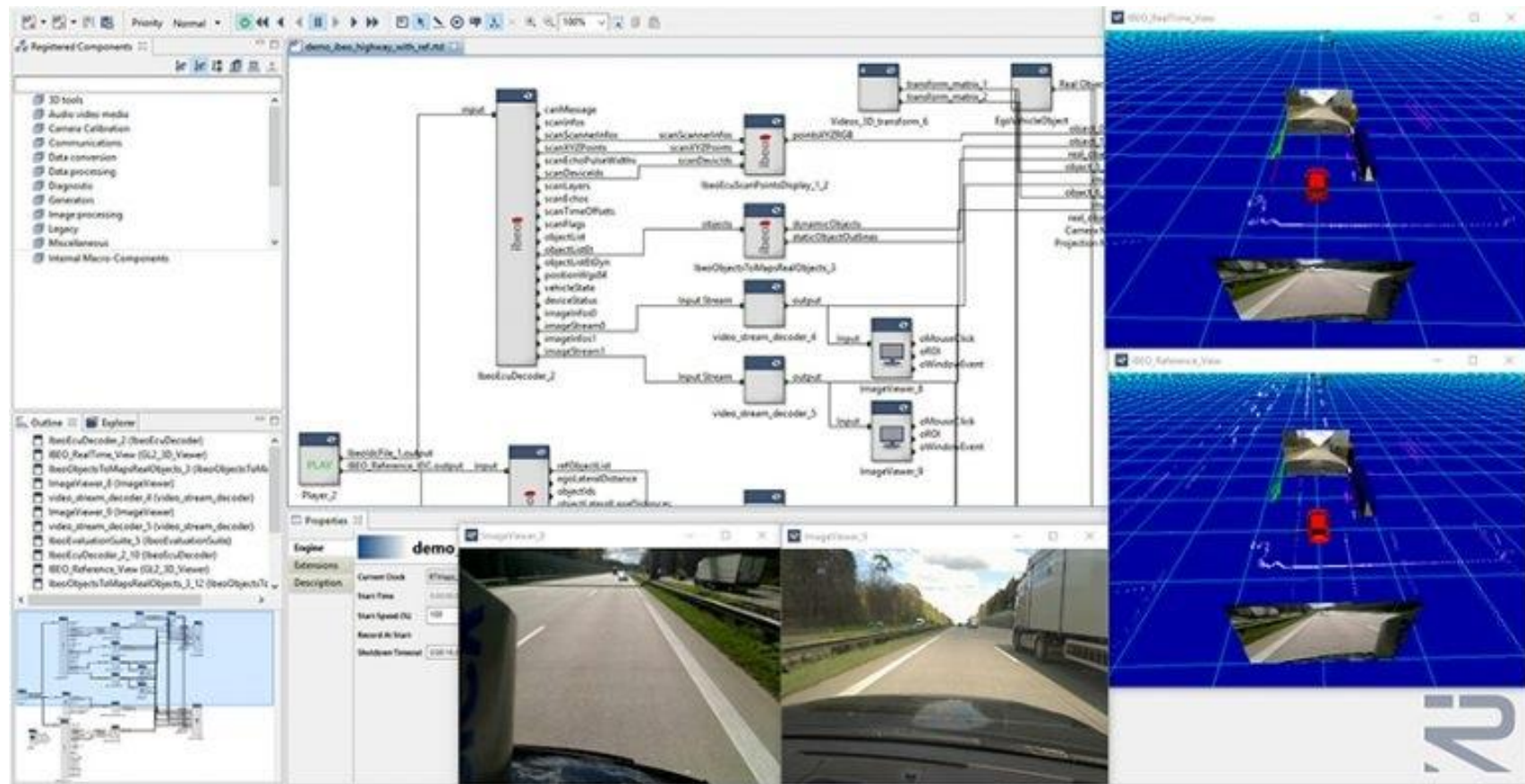
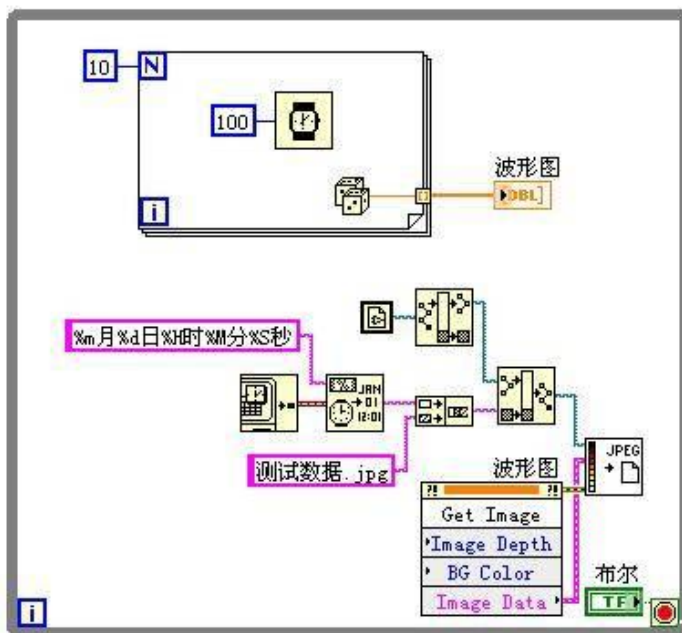
Examples of Direct Manipulation

- Computer-Aided Design (CAD) / Manufacturing (CAM)
 - AutoCAD
 - 3D Max
- Digital Twins



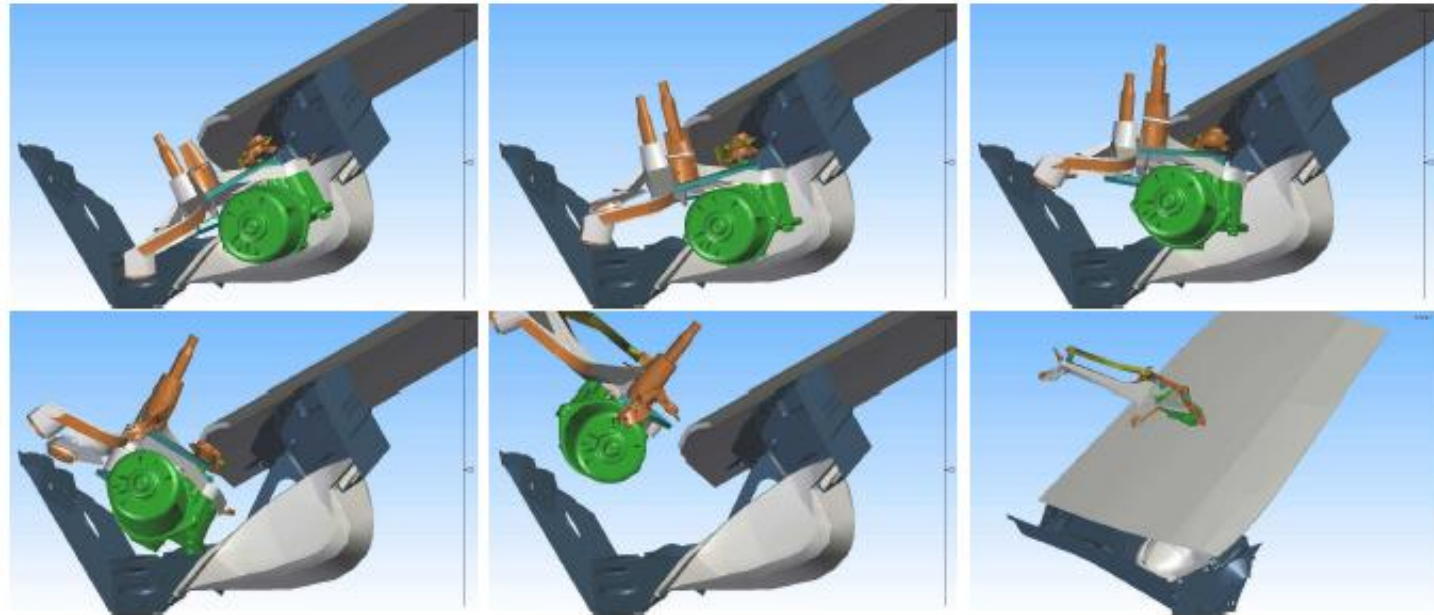
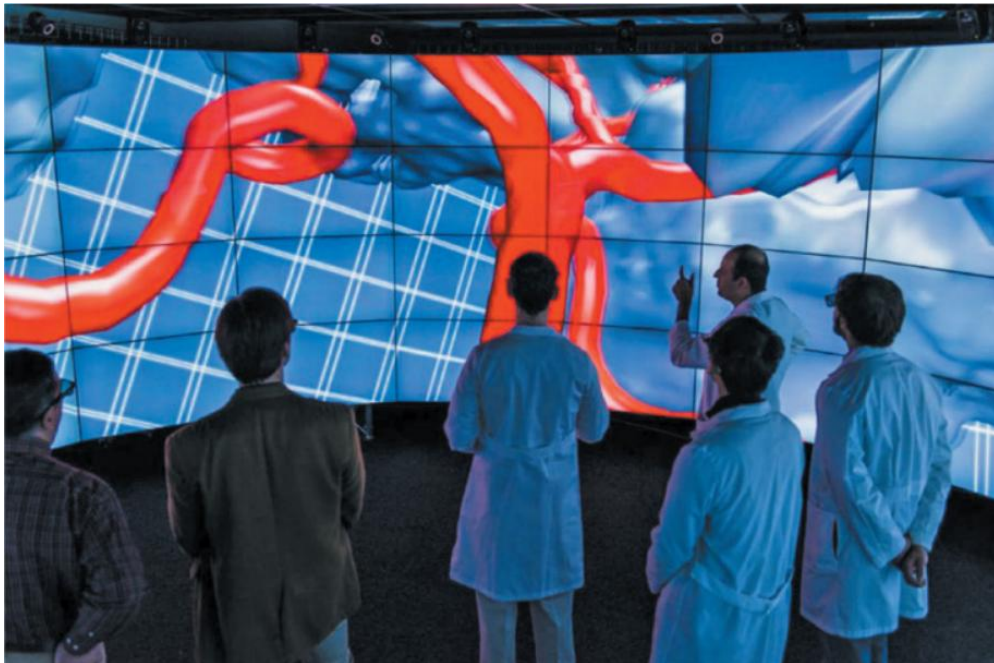
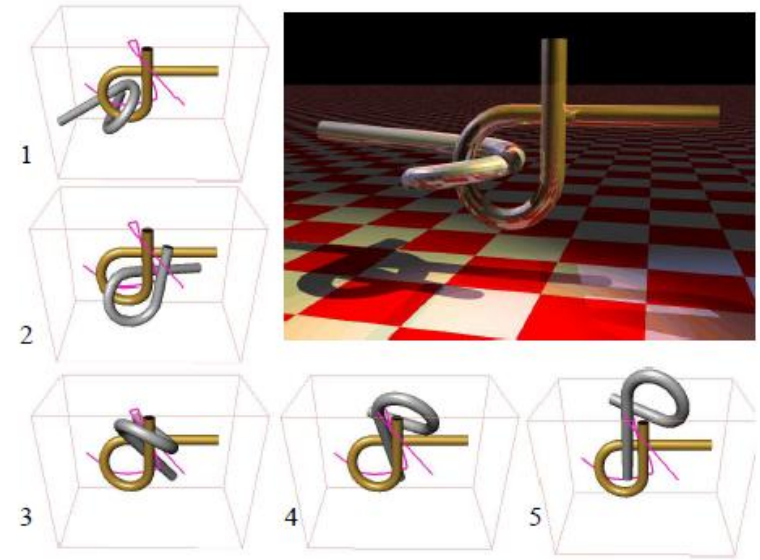
- Direct-Manipulated Programming / Configuration

- Excel
- Labview
- RTMaps



Examples of Direct Manipulation

- Three-Dimensional (3D) UI
 - Handling medical difficulty
 - Handling industrial difficulty



Examples of Direct Manipulation

- Tele / Remote Direct Manipulation
 - Teleoperation
 - Mobile remote presence (MRP)

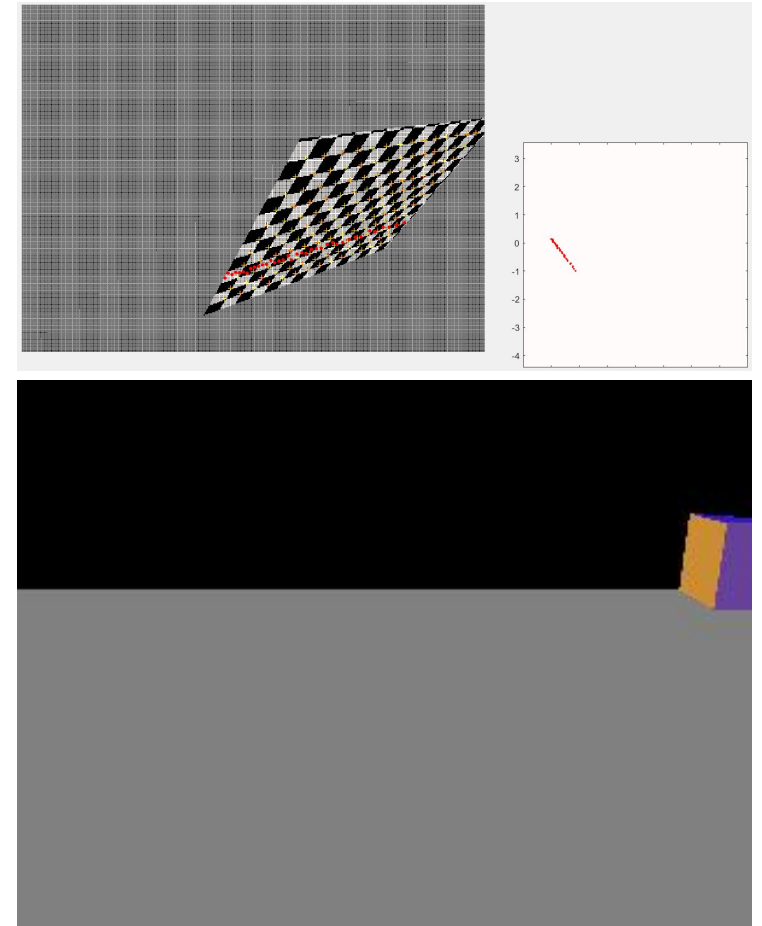


Virtual Reality and Augmented Reality

- Virtual Reality
 - Refers to generating completely synthetic images that simulate the real world as well as purely imagined reality
- Augmented Reality
 - Refers to augmenting images actually captured from the real world by virtual elements to facilitate conception of the images

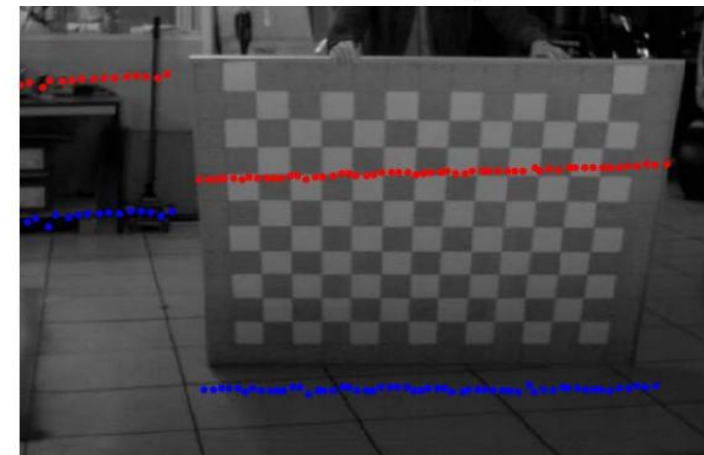
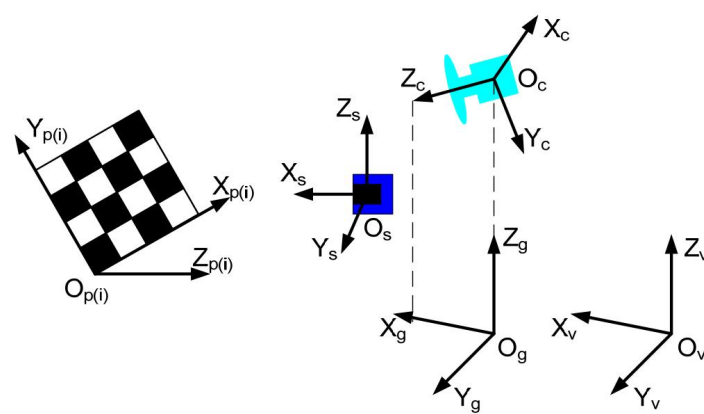
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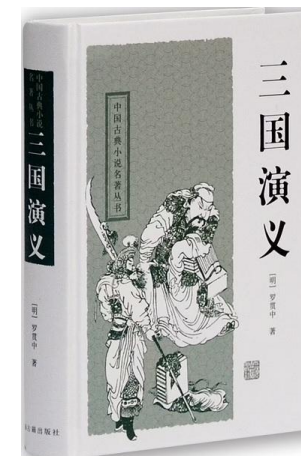
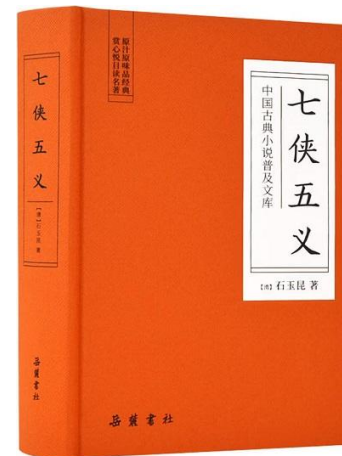
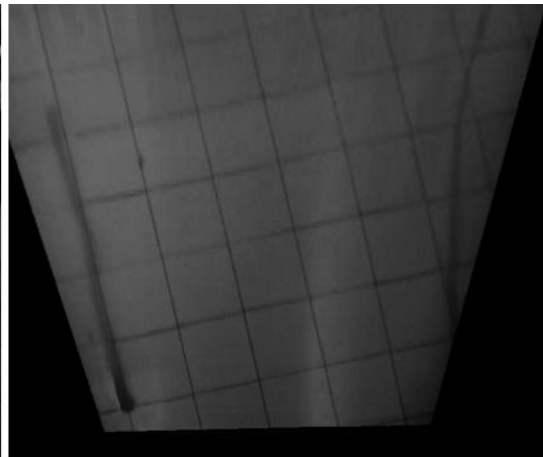
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Virtual Reality and Augmented Reality

- Virtual Reality vs. Augmented Reality
 - VR refers to generating completely synthetic images that simulate the real world as well as purely imagined reality
 - AR refers to augmenting images actually captured from the real world by virtual elements to facilitate conception of the images
- Practice
 - Is it VR or AR?



DISCUSSION

Q & A